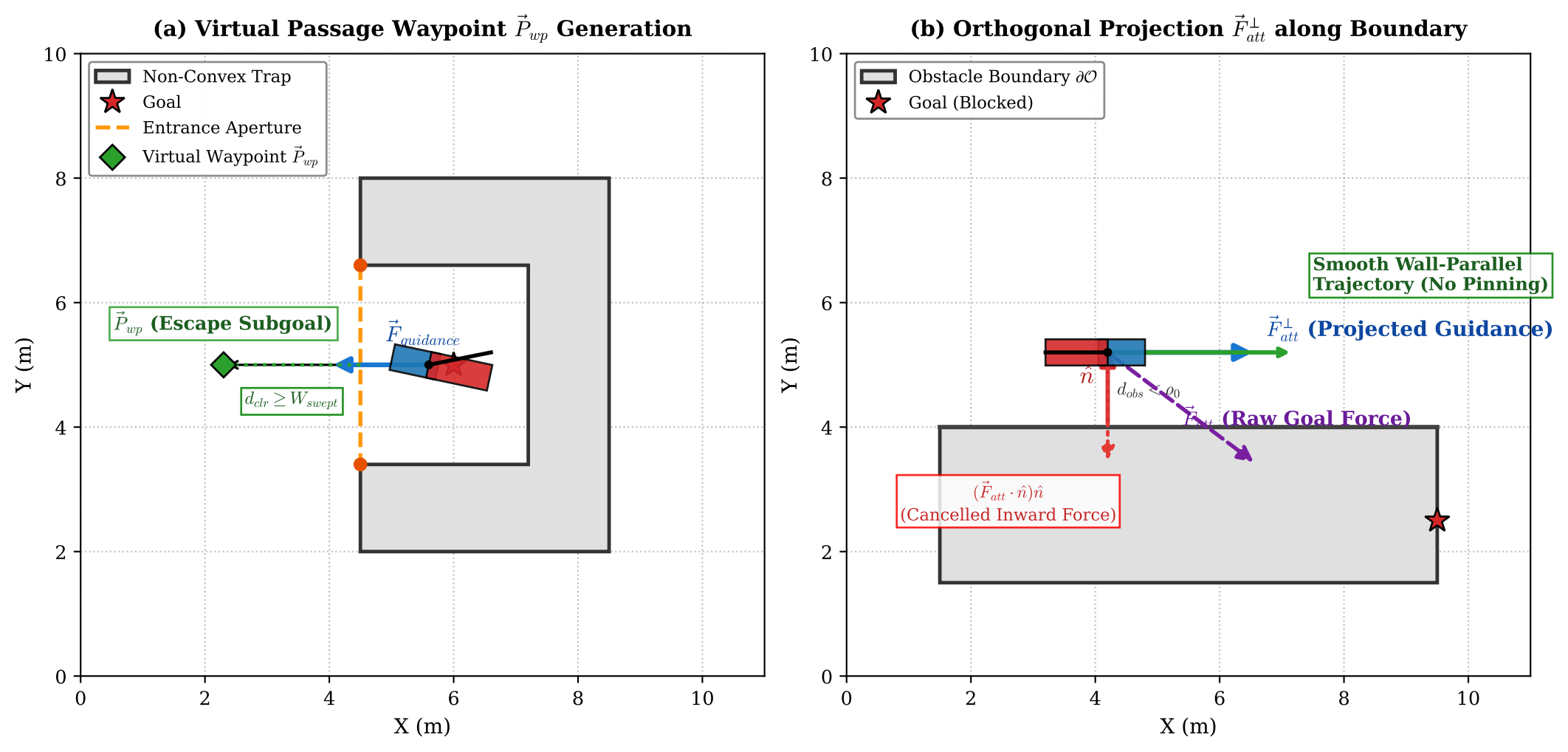

(a) Virtual Passage Waypoint $\vec{P}_{wp}$ Generation
Non-Convex Trap
Goal
Entrance Aperture
Virtual Waypoint $\vec{P}_{wp}$
$\vec{P}_{wp}$ (Escape Subgoal)
$\vec{F}_{guidance}$
$d_{clr} \geq W_{swept}$
Y (m)
X (m)
(b) Orthogonal Projection $\vec{F}_{att}^{\perp}$ along Boundary
Obstacle Boundary $\partial\mathcal{O}$
Goal (Blocked)
Smooth Wall-Parallel Trajectory (No Pinning)
$\vec{F}_{att}^{\perp}$ (Projected Guidance)
$\hat{n}$
$d_{obs} < \rho_0$
$\vec{F}_{att}$ (Raw Goal Force)
$(\vec{F}_{att} \cdot \hat{n})\hat{n}$
(Cancelled Inward Force)
Y (m)
X (m)